

Semi-Active Suspension Control Development

A Project Report Submitted For The Company
UNO MINDA



by

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Abstract

Ride comfort and **vibration isolation** are critical in the design of vehicle seat suspension systems, especially for prolonged operator use. Here in first phase of project, we focus on developing a seat suspension model, aiming to accurately capture the dynamic response under various road excitations.

A **mathematical model of the seat suspension system** was formulated based on the principles of mass-spring-damper, incorporating the effects of seat mass, suspension stiffness, and damping. MATLAB was used to simulate system behavior for different operating conditions and perform **system identification**.

We then use this baseline model for evaluating ride comfort metrics and the development of control strategy. We have tuned our controller parameters that uses a **Skyhook control law** for regulating the viscosity of **MR (Magnetorheological) damper** by plotting frequency responses. We keep the best set of chosen parameters from the frequency response with lowest resonant peak

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Contents

1	Introduction	6
2	System Modeling	8
2.1	System Overview	8
2.2	Mathematical Formulation	8
2.3	Geometrical Relationships	9
2.4	Air Spring Model	9
2.5	Damper Model	10
2.6	MATLAB Implementation	10
2.7	Model Validation Approach	11
3	Data Processing and Experimental Setup	12
3.1	Company Provided Data Overview	12
3.2	Data Filtering Techniques	12
3.3	Unfiltered and Filtered Data Comparison	13
3.4	Frequency Response Analysis	14
3.5	Amplification Analysis	14
3.6	Comparative Analysis of Different Load Conditions	15
3.7	Implications for Model Development	15
4	Geometrical and Analytical Modeling	16
4.1	Geometrical Configuration of the Air Spring	16
4.2	Geometrical Configuration of the Damper	17
4.3	MATLAB Implementation of Geometrical Models	18
4.4	Simulation with Tilted Spring and Constant Equivalent Damper	18
4.5	Finding the Backbone Spring Force Curves	18
4.6	Incorporating Geometrical Effects into System Dynamics	19
4.7	Validation of Geometrical Models	19
5	Simulation and Analysis	20
5.1	Linear Approximation at 3 Bar Pressure	20
5.2	Simulation with Inclined Spring and Damper	20
5.3	Analysis of Discrepancies	21
5.4	Simulation with Auxiliary Damper in Parallel to Air Spring Axis	22
5.5	Parametric Study of Spring Stiffness and Damping	23
5.6	Frequency Response with Progressive Damper	23
5.7	Implications for Control System Design	23

6	Component-wise Evaluation	25
6.1	Component-wise Testing of Air Spring	25
6.2	Modeling Effective Air Spring Force	25
6.3	Non-linear Backbone Curve of Stiffness Factor	26
6.4	Experimentation with Auxiliary Damping Coefficient	26
6.5	Simulation of Air Spring Curves	26
6.6	Simulation with 4-Bar Pressure Curve	28
6.7	Mathematical Explanation of Hysteresis Modeling	28
6.8	Validation of Component Model	29
7	Final Integrated Results	30
7.1	Combined Model Configuration	30
7.2	Frequency Response After Combining All Components	30
7.3	Analysis of Model Accuracy	31
7.4	Comparison of all Models	31
7.5	Remaining Discrepancies and Potential Sources of Error	33
8	Control system design	36
8.1	Implications for Control System Design	36
8.2	Semi-active Seat Suspension Model and its Dynamics	36
8.3	General working zone of suspension systems	37
8.4	Selection Rationale of our Control law	38
8.5	Improvised Skyhook control law	38
8.6	SIMULATION RESULTS	39
9	Comparison of our Seat Suspension Systems with SEARS seat	42
10	Differentiation from the US Patent Seat	46
11	Conclusion and Future Work	48

List of Figures

1	Filtering data for 80 kg load condition	13
2	Frequency response curve for 40 and 80 kg load condition	13
3	Geometrical configuration profile of the air spring	16
4	Geometrical configuration profile of the damper	17
5	Fitting the backbone curves by digitization	19
6	Frequency response simulation with inclined spring and PR damper	21
7	Frequency response : simulation with auxiliary damper(BEST RESULTS)	22
8	Hsteresis width calculation	27

9	Simulated air spring force-displacement curves at different frequencies at damping coefficients	27
10	Best matching result at .4 hertz	28
11	Frequency response curve for the final model	31
12	Frequency response simulation with experimental data for 80 kg(company given)	32
13	Frequency response simulation with inclined spring and PR damper . . .	32
14	Frequency response : simulation with auxiliary damper(BEST RESULTS)	33
15	Frequency response curve for the final model	33
16	Semi-active Seat Suspension System Model diagram	36
17	Different suspension scheme control regions	37
18	Comparision of the seat suspension system performance with PR damper , and the MR damper(with skyhook control) for an 80kg passenger . . .	39
19	Comparision of the seat suspension system performance with PR damper , and the MR damper(with skyhook control) for an 40kg passenger . . .	40
20	CONTROL LOGIC FLOW FOR ACTIVE, SEMIACTIVE AND PAS-SIVE MODES	47

List of Tables

1	Comparison of model configurations with experimental data	32
2	Comparison of UNO MINDA and SEARS Seat Suspension Systems . . .	42

1 Introduction

Vehicle suspension systems play a crucial role in ensuring ride comfort, handling, and stability. Among the various types of suspension systems, semi-active suspensions offer a compelling compromise between the simplicity of passive systems and the performance of active systems. This thesis focuses on the development of a semi-active seat suspension system, specifically designed to enhance ride comfort and vibration isolation for operators subjected to prolonged periods of vehicle operation. The importance of effective seat suspension cannot be overstated, particularly in commercial vehicles, heavy machinery, and long-haul transportation where operators remain seated for extended durations. Poor vibration isolation can lead to driver fatigue, reduced productivity, and long-term health issues. Therefore, optimizing seat suspension systems is not merely a matter of comfort but also a significant factor in operational safety and efficiency. This research represents the first phase of a comprehensive project aimed at developing an advanced semi-active seat suspension system. The primary objective is to create an accurate mathematical model that captures the dynamic behavior of the seat suspension under various road excitation conditions. This model will serve as the foundation for subsequent phases involving optimization and implementation of sophisticated control strategies. The scope of this work encompasses several key aspects. Firstly, it involves the formulation of a mathematical model based on mass-spring-damper principles, accounting for the seat mass, suspension stiffness, and damping characteristics. Secondly, it includes the processing and analysis of experimental data provided by industry partners. Thirdly, it involves the development of geometrical models for critical components such as the air spring and damper. Finally, it encompasses extensive simulation work to validate the model against experimental results and identify areas for improvement. The methodology employed in this research combines theoretical modeling with practical validation. MATLAB serves as the primary simulation environment, enabling the analysis of system behavior under various operating conditions. The modeling approach incorporates both linear and non-linear elements to accurately represent the complex dynamics of the suspension system. The significance of this work lies in its potential to contribute to the advancement of seat suspension technology. By developing an accurate baseline model, this research paves the way for the implementation of semi active control strategy that can adapt to changing road conditions and operator requirements. The ultimate future goal is to replace the PR damper with an MR damper that can be magnetically controlled for superior ride comfort and fully automated optimal seat suspension system.

FIRST PHASE REPORT

2 System Modeling

The mathematical modeling of the seat suspension system forms the foundation of this research. This chapter details the formulation of the mathematical model based on mass-spring-damper principles, incorporating the effects of seat mass, suspension stiffness, and damping characteristics. The model aims to accurately capture the dynamic response of the seat suspension under various road excitation conditions.

2.1 System Overview

The seat suspension system under consideration consists of several key components: the seat mass, an air spring providing variable stiffness, a progressive damper offering variable damping characteristics, and the connecting linkage mechanisms. The system is designed to isolate the operator from vibrations transmitted through the vehicle chassis while maintaining appropriate seat height and stability. The air spring operates with a pressure range of 1 to 7 bar, allowing for adjustment of the spring characteristics based on the operator's weight and desired ride comfort. The progressive damper provides damping forces that vary with the velocity of the suspension motion, offering improved control over the suspension dynamics compared to linear dampers. The suspension mechanism incorporates a geometrical configuration that results in non-linear kinematic relationships between the seat displacement and the spring/damper displacements. This geometrical arrangement plays a crucial role in determining the overall dynamic behavior of the system.

2.2 Mathematical Formulation

The seat suspension system is modeled as a single-degree-of-freedom system with the seat mass representing the sprung mass. The equation of motion for the system can be derived using Newton's second law:

$$m_s \ddot{z}_s = F_{spring} + F_{damper} + F_{friction} + F_{input} \quad (1)$$

where: - m_s is the seat mass (kg) - z_s is the vertical displacement of the seat (m) - F_{spring} is the force exerted by the air spring (N) - F_{damper} is the force exerted by the damper (N) - $F_{friction}$ represents friction forces in the system (N) - F_{input} represents external input forces (N) The air spring force is modeled as a non-linear function of the spring displacement and velocity:

$$F_{spring} = f_k(x_{as}, \dot{x}_{as}) \quad (2)$$

where x_{as} and $\dot{x}_{as}$ are the displacement and velocity of the air spring, respectively. The function f_k captures the non-linear stiffness characteristics of the air spring, including

hysteresis effects. The damper force is modeled as a function of the damper velocity:

$$F_{damper} = f_c(\dot{x}_d) \quad (3)$$

where $\dot{x}_d$ is the velocity of the damper. The function f_c represents the progressive damping characteristics, which can vary significantly with velocity. The friction forces are modeled using a simplified Coulomb friction model:

$$F_{friction} = -F_c \cdot \text{sign}(\dot{z}_s) \quad (4)$$

where F_c is the Coulomb friction coefficient.

2.3 Geometrical Relationships

The geometrical configuration of the suspension mechanism introduces non-linear relationships between the seat displacement and the spring/damper displacements. These relationships are crucial for accurately modeling the system dynamics. The spring length l_{as} as a function of seat height h_s is given by:

$$l_{as} = f_{l_{as}}(h_s) \quad (5)$$

Similarly, the spring angle ζ with respect to the ground is given by:

$$\zeta = f_{\zeta}(h_s) \quad (6)$$

For the damper, the length l_d and angle β with respect to the ground are given by:

$$l_d = f_{l_d}(h_s) \quad (7)$$

$$\beta = f_{\beta}(h_s) \quad (8)$$

These geometrical relationships are implemented in MATLAB as functions that return the respective values based on the seat height. The detailed implementation of these functions is provided in Chapter 4.

2.4 Air Spring Model

The air spring model is a critical component of the overall system model. Air springs exhibit complex non-linear behavior due to the thermodynamic properties of the compressed air and the material characteristics of the bellows. The force generated by the air spring can be expressed as:

$$F_{spring} = (P - P_{atm}) \cdot A_{eff} \quad (9)$$

where: - P is the absolute pressure inside the air spring (Pa) - P_{atm} is the atmospheric pressure (Pa) - A_{eff} is the effective area of the air spring (m^2) The pressure inside the

air spring follows the polytropic process:

$$P \cdot V^n = \text{constant} \quad (10)$$

where: - V is the volume of the air spring (m^3) - n is the polytropic index (typically between 1.0 and 1.4) The effective area A_{eff} and the volume V are functions of the spring displacement, introducing non-linearities into the model. These relationships are typically determined experimentally or through detailed computational fluid dynamics (CFD) analysis. In addition to the primary spring force, air springs exhibit hysteresis behavior due to the viscoelastic properties of the bellows material and the thermodynamic effects of air compression and expansion. This hysteresis is modeled using an auxiliary damping component in parallel with the spring:

$$F_{spring} = F_{elastic} + F_{hysteresis} \quad (11)$$

where $F_{elastic}$ represents the elastic component of the spring force, and $F_{hysteresis}$ represents the hysteresis component, modeled as:

$$F_{hysteresis} = c_{aux} \cdot \dot{x}_{as} \quad (12)$$

where c_{aux} is the auxiliary damping coefficient.

2.5 Damper Model

The progressive damper model captures the non-linear damping characteristics that vary with the velocity of the damper motion. The damping force is modeled as:

$$F_{damper} = c(\dot{x}_d) \cdot \dot{x}_d \quad (13)$$

where $c(\dot{x}_d)$ is the variable damping coefficient, which is a function of the damper velocity. For a progressive damper, the damping coefficient typically increases with velocity, providing higher damping forces at higher velocities. This behavior can be modeled using a piecewise function or a continuous non-linear function. In this research, the progressive damper is modeled using a detailed geometrical representation that accounts for the orientation and mounting configuration of the damper in the suspension system. The damping characteristics are derived from experimental data provided by the manufacturer.

2.6 MATLAB Implementation

The mathematical model is implemented in MATLAB, leveraging its powerful numerical computation and simulation capabilities. The implementation consists of several key components: 1. Main simulation script that defines the system parameters and runs the simulation 2. Function for computing the geometrical relationships (spring length, spring angle, damper length, damper angle) 3. Function for computing the air spring

force, including hysteresis effects 4. Function for computing the damper force based on the progressive damping characteristics 5. Numerical integration routine for solving the equations of motion The simulation allows for the analysis of the system response under various operating conditions, including different loads, road excitation profiles, and system parameters. The results of these simulations are presented and discussed in subsequent chapters.

2.7 Model Validation Approach

The validation of the mathematical model is a critical aspect of this research. The model validation approach involves comparing the simulation results with experimental data provided by the company. Key metrics for model validation include: - Natural frequency of the system - Resonance peak amplitude - Frequency response characteristics - Transient response to step inputs The validation process involves an iterative approach, where model parameters are adjusted to minimize the discrepancy between simulation and experimental results. This process is detailed in subsequent chapters, particularly in Chapter 6 and Chapter 7.

3 Data Processing and Experimental Setup

The foundation of any robust suspension model lies in the quality and proper processing of experimental data. This chapter details the methodologies employed in processing and analyzing the company-provided data for the seat suspension system. The data processing pipeline involved several stages, including initial data assessment, filtering techniques, and frequency response analysis, all of which were critical for establishing an accurate baseline model.

3.1 Company Provided Data Overview

The experimental data provided by UNO MINDA comprised frequency response measurements for the seat suspension system under different loading conditions. Specifically, data sets were provided for 40 kg and 80 kg loads, representing the range of operator weights the system is designed to accommodate. These data sets included both unfiltered and filtered frequency response curves, which served as benchmarks for model validation. The frequency response data included measurements of transmissibility (the ratio of output amplitude to input amplitude) across a range of excitation frequencies. This data is particularly valuable as it directly relates to ride comfort metrics, with lower transmissibility values indicating better vibration isolation.

3.2 Data Filtering Techniques

Raw experimental data often contains noise and artifacts that can obscure the underlying system dynamics. To address this, a Butterworth filter was applied to the raw data to extract the meaningful frequency response characteristics. The Butterworth filter is a type of signal processing filter designed to have a frequency response as flat as possible in the passband. For this application, a first-order Butterworth filter with a bandwidth of 0.4 Hz was employed. The center frequencies ranged from 0.5 Hz to 2.0 Hz with intervals of 0.1 Hz, allowing for detailed analysis across the critical frequency range for seat suspension systems. The mathematical representation of a first-order Butterworth low-pass filter is given by:

$$|H(j\omega)|^2 = \frac{1}{1 + \left(\frac{\omega}{\omega_c}\right)^{2n}} \quad (14)$$

where: - $H(j\omega)$ is the transfer function - ω is the angular frequency - ω_c is the cutoff frequency - n is the filter order (1 in this case) The filtering process was implemented in MATLAB, utilizing the built-in signal processing functions. The filtered data provided a clearer representation of the system's frequency response, facilitating more accurate model validation.

3.3 Unfiltered and Filtered Data Comparison

A comparative analysis of unfiltered and filtered data revealed significant differences in the clarity and interpretability of the frequency response curves. The unfiltered data exhibited considerable noise, particularly at higher frequencies, which could lead to misinterpretation of the system's dynamic characteristics. Figure 1 presents the unfiltered frequency response data for the 80 kg load condition. As can be observed, the curve shows significant fluctuations, making it difficult to identify the true resonance frequency and peak amplitude.

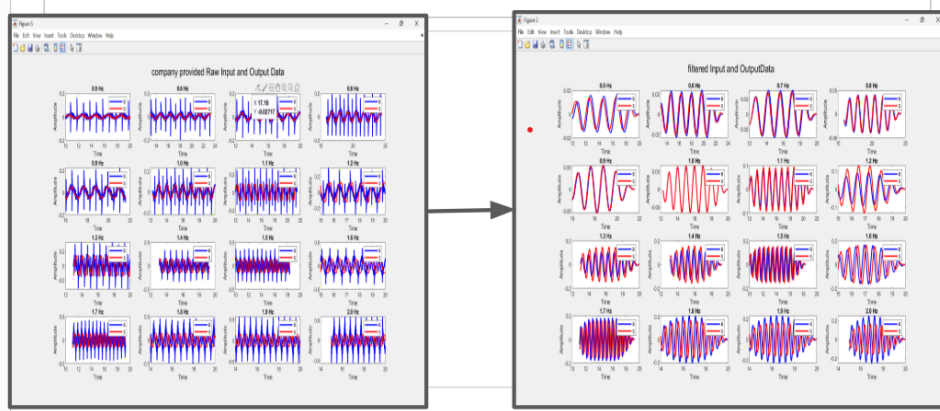


Figure 1: Filtering data for 80 kg load condition

In contrast, Figure 2 shows the same data after applying the Butterworth filter. The filtered curve exhibits a smoother profile, with a clearly identifiable resonance peak at approximately 1.2 Hz and a peak amplification of 1.42.

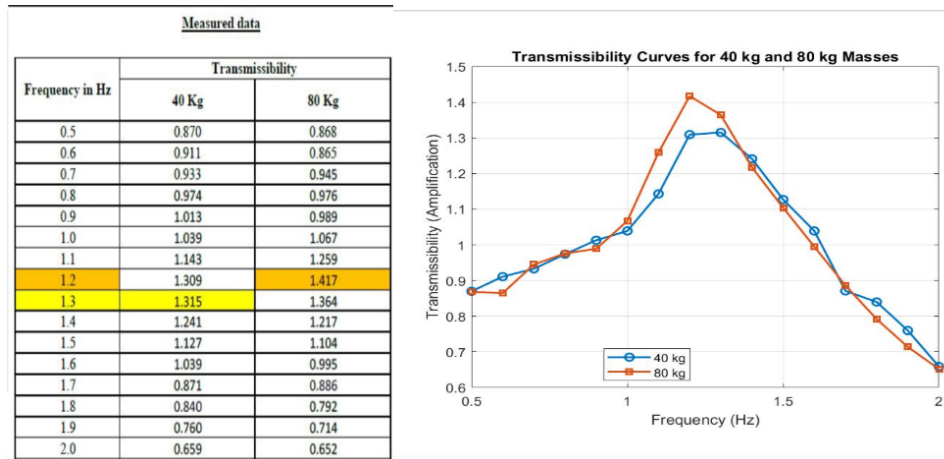


Figure 2: Frequency response curve for 40 and 80 kg load condition

The filtering process not only improved the clarity of the frequency response curves but also revealed subtle features of the system dynamics that were obscured by noise in the unfiltered data. This was particularly important for identifying the resonance characteristics and for subsequent model validation.

3.4 Frequency Response Analysis

The filtered frequency response data provided valuable insights into the dynamic characteristics of the seat suspension system. For the 80 kg load condition, the system exhibited a resonance frequency of approximately 1.2 Hz with a peak amplification of 1.42. This resonance frequency is within the typical range for seat suspension systems and corresponds to the natural frequency of the system under this loading condition. The frequency response curve also revealed the system's behavior across a range of frequencies. At frequencies well below the resonance frequency (below 0.5 Hz), the transmissibility was close to unity, indicating that the system effectively transmits low-frequency inputs without significant amplification or attenuation. At frequencies well above the resonance frequency (above 2.0 Hz), the transmissibility decreased, indicating that the system provides effective vibration isolation at higher frequencies. For the 40 kg load condition, the frequency response characteristics were different, reflecting the change in system dynamics due to the reduced mass. The resonance frequency for this condition was slightly higher than that for the 80 kg load, which is consistent with the theoretical relationship between natural frequency and mass:

$$f_n = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \quad (15)$$

where: - f_n is the natural frequency (Hz) - k is the system stiffness (N/m) - m is the mass (kg) This relationship indicates that for a constant stiffness, a reduction in mass results in an increase in natural frequency, which was observed in the experimental data.

3.5 Amplification Analysis

The amplification plot, which shows the ratio of output amplitude to input amplitude as a function of frequency, provided additional insights into the system's dynamic characteristics. The peak amplification value is a critical parameter for assessing ride comfort, as higher amplification at resonance can lead to discomfort for the operator. For the 80 kg load condition, the peak amplification was 1.42, occurring at the resonance frequency of 1.2 Hz. This value represents a moderate level of amplification, indicating that the system provides reasonable vibration isolation at resonance. However, there is room for improvement, particularly through the implementation of semi-active control strategies that can adaptively adjust the damping characteristics to reduce resonance amplification. The amplification plot also revealed the bandwidth of effective vibration isolation. Frequencies above approximately 2.0 Hz showed amplification values less than 1.0, indicating that the system effectively attenuates vibrations in this frequency range. This is particularly important for ride comfort, as many road-induced vibrations occur at frequencies above 2.0 Hz.

3.6 Comparative Analysis of Different Load Conditions

A comparative analysis of the frequency response characteristics under different load conditions (40 kg and 80 kg) revealed important insights into the system's behavior. As expected, the resonance frequency was higher for the 40 kg load condition compared to the 80 kg load condition, reflecting the inverse relationship between natural frequency and mass. However, the peak amplification was similar for both load conditions, indicating that the damping characteristics of the system are relatively consistent across the range of operating loads. This is a desirable characteristic, as it suggests that the system can provide consistent ride comfort for operators of different weights. The comparative analysis also revealed differences in the roll-off characteristics (the rate at which amplification decreases with increasing frequency) between the two load conditions. The 80 kg load condition exhibited a steeper roll-off, indicating better high-frequency vibration isolation for heavier operators. This difference can be attributed to the changes in the system's dynamic characteristics due to the varying load.

3.7 Implications for Model Development

The data processing and analysis phase provided critical insights that informed the subsequent model development. The identification of resonance frequencies and peak amplification values served as key validation metrics for the mathematical model. The filtering process highlighted the importance of proper data preprocessing in extracting meaningful system dynamics from experimental data. This understanding was incorporated into the model validation process, where similar filtering techniques were applied to simulation results for fair comparison with experimental data. The comparative analysis of different load conditions underscored the need for a model that can accurately capture the system's behavior across the range of operating conditions. This led to the incorporation of load-dependent parameters in the mathematical model, particularly for the air spring stiffness characteristics.

4 Geometrical and Analytical Modeling

The geometrical configuration of the seat suspension system plays a crucial role in determining its dynamic behavior. This chapter details the development of the geometrical models for the air spring and damper components, which are essential for accurately capturing the kinematic relationships within the suspension system.

4.1 Geometrical Configuration of the Air Spring

The air spring in the seat suspension system is mounted in an inclined orientation, which introduces non-linear kinematic relationships between the seat displacement and the spring compression/extension. This configuration is designed to optimize the packaging and performance characteristics of the suspension system. The geometrical configuration of the air spring is characterized by two key parameters as functions of seat height: the spring length and the spring angle with respect to the ground. These relationships are derived from the physical dimensions and mounting points of the suspension system. Figure 3 illustrates the geometrical configuration of the air spring, showing the key dimensions and angles that define its orientation in the suspension system.

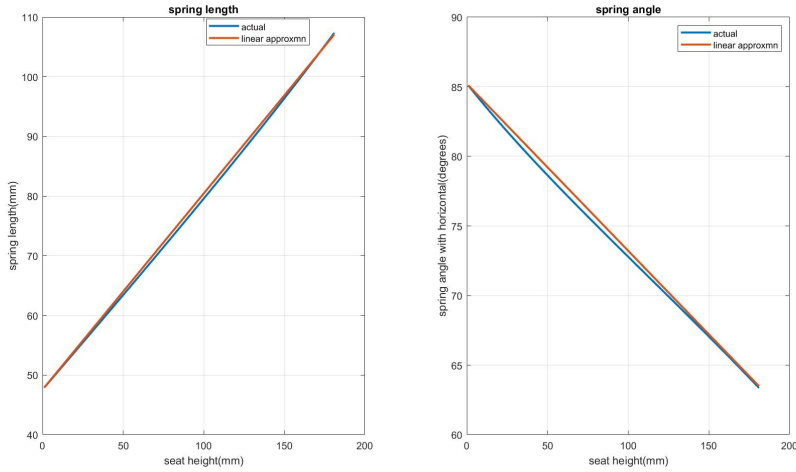


Figure 3: Geometrical configuration profile of the air spring

The spring length l_{as} as a function of seat height h_s is given by the equation:

$$l_{as} = \sqrt{(x_{m2} - x_{m1})^2 + (y_{m2} - y_{m1})^2} \quad (16)$$

where (x_{m1}, y_{m1}) and (x_{m2}, y_{m2}) are the coordinates of the mounting points of the air spring, which are functions of the seat height h_s . Similarly, the spring angle ζ with

respect to the ground is given by:

$$\zeta = \arctan \left(\frac{y_{m2} - y_{m1}}{x_{m2} - x_{m1}} \right) \quad (17)$$

These relationships are implemented in MATLAB as functions that return the spring length and angle based on the seat height. The implementation accounts for the specific mounting points and linkage geometry of the suspension system.

4.2 Geometrical Configuration of the Damper

Similar to the air spring, the damper is also mounted in an inclined orientation, which introduces additional non-linear kinematic relationships. The damper's geometrical configuration is characterized by its length and angle with respect to the ground as functions of seat height. Figure 4 illustrates the geometrical configuration of the damper, showing the key dimensions and angles that define its orientation in the suspension system.

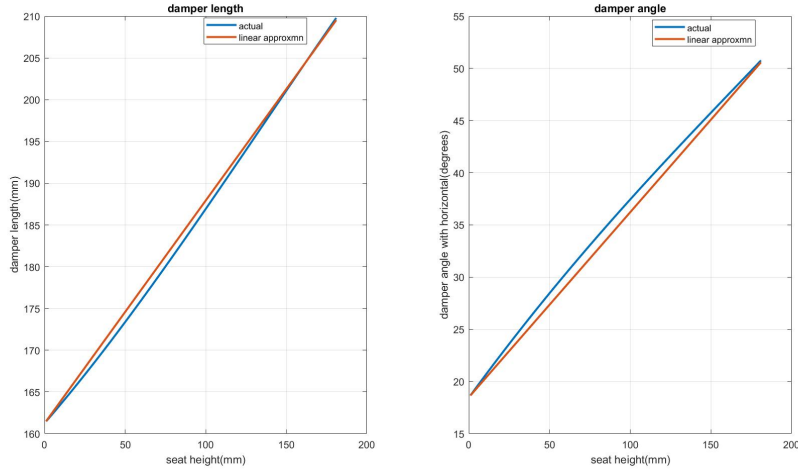


Figure 4: Geometrical configuration profile of the damper

The damper length l_d as a function of seat height h_s is given by the equation:

$$l_d = \sqrt{(x_{m4} - x_{m3})^2 + (y_{m4} - y_{m3})^2} \quad (18)$$

where (x_{m3}, y_{m3}) and (x_{m4}, y_{m4}) are the coordinates of the mounting points of the damper, which are functions of the seat height h_s . The damper angle β with respect to the ground is given by:

$$\beta = \arctan \left(\frac{y_{m4} - y_{m3}}{x_{m4} - x_{m3}} \right) \quad (19)$$

These relationships are also implemented in MATLAB as functions that return the damper length and angle based on the seat height.

4.3 MATLAB Implementation of Geometrical Models

The geometrical models for the air spring and damper are implemented in MATLAB as a set of functions that calculate the key geometrical parameters based on the seat height. The implementation is contained in a MATLAB file named "seat_geom_model.m", which creates four functions: 1. 'f_Las': Returns the spring length as a function of seat height 2. 'f_zeta': Returns the spring angle with respect to the ground as a function of seat height 3. 'f_Ld': Returns the damper length as a function of seat height 4. 'f_beta': Returns the damper angle with respect to the ground as a function of seat height These functions are essential for accurately modeling the kinematic relationships within the suspension system, which in turn affect the dynamic behavior of the system. The implementation accounts for the specific mounting points and linkage geometry of the suspension system, ensuring that the geometrical relationships are accurately captured.

4.4 Simulation with Tilted Spring and Constant Equivalent Damper

The geometrical models were used to simulate the behavior of the suspension system with a tilted spring and a constant equivalent damper. This simulation aimed to validate the geometrical models and to understand the effects of the inclined orientation on the system dynamics. For the simulation, a nominal stiffness value of $k = 63.5 \times 10^3$ N/m was used for the air spring at 4 bar pressure, which corresponds to the 80 kg mass condition. This value was derived from experimental data provided by the company. The damping coefficient value of $c = 4800$ Ns/m was used for the damper, as this value was shown to bring the resonance around 1.2 Hz experimentally. This value represents the equivalent damping coefficient that accounts for the progressive damping characteristics of the actual damper. The simulation results showed that the inclined orientation of the spring and damper introduces non-linear kinematic relationships that significantly affect the system dynamics. The effective stiffness and damping coefficients vary with the seat height, leading to complex dynamic behavior that cannot be captured by simple linear models.

4.5 Finding the Backbone Spring Force Curves

The backbone spring force curves represent the relationship between the spring force and displacement, accounting for the non-linear characteristics of the air spring. These curves are essential for accurately modeling the spring behavior, particularly the hysteresis effects. To find the backbone spring force curves, experimental data was used to determine the force-displacement relationship at different pressures. The nominal pressure in the air spring for the 80 kg experiment was 4 bar, which served as a reference point for the analysis. The backbone spring force curves were determined by analyzing the experimental data and fitting appropriate mathematical functions to capture the non-linear characteristics. These curves were then incorporated into the mathematical model to accurately represent the spring behavior. The analysis revealed that the air spring exhibits signifi-

cant non-linear behavior, with the stiffness varying with displacement and pressure. This non-linearity is essential to capture in the model, as it significantly affects the system dynamics, particularly the resonance characteristics.

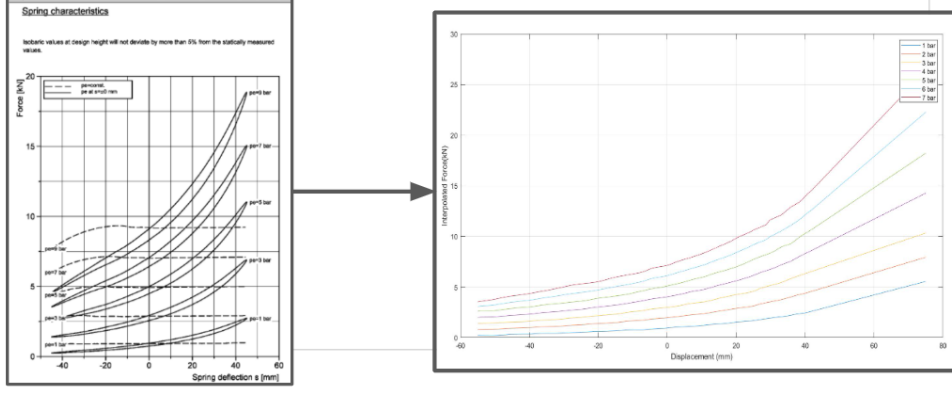


Figure 5: Fitting the backbone curves by digitization

4.6 Incorporating Geometrical Effects into System Dynamics

The geometrical effects of the inclined spring and damper orientations were incorporated into the system dynamics by transforming the spring and damper forces into the vertical direction. This transformation accounts for the angles of the spring and damper with respect to the ground, ensuring that the forces are properly resolved in the direction of seat motion. The vertical component of the spring force is given by:

$$F_{spring,vertical} = F_{spring} \cdot \sin(\zeta) \quad (20)$$

where F_{spring} is the force exerted by the spring, and ζ is the spring angle with respect to the ground. Similarly, the vertical component of the damper force is given by:

$$F_{damper,vertical} = F_{damper} \cdot \sin(\beta) \quad (21)$$

where F_{damper} is the force exerted by the damper, and β is the damper angle with respect to the ground. These transformations are essential for accurately modeling the system dynamics, as they account for the geometrical configuration of the suspension system. Without these transformations, the model would not accurately capture the dynamic behavior of the system, particularly the resonance characteristics.

4.7 Validation of Geometrical Models

The geometrical models were validated by comparing the simulated system response with experimental data. The validation focused on key metrics such as resonance frequency and peak amplification, which are sensitive to the geometrical configuration of the system. The validation results showed that the geometrical models accurately capture the kinematic relationships within the suspension system.

5 Simulation and Analysis

Frequency response analysis is a fundamental tool for understanding the dynamic behavior of suspension systems. This chapter presents a comprehensive analysis of the frequency response characteristics of the seat suspension system under various combinations of springs and dampers. The analysis provides critical insights into the system's performance and serves as a basis for model validation and optimization.

5.1 Linear Approximation at 3 Bar Pressure

As a starting point for the frequency response analysis, a linear approximation of the air spring behavior at 3 bar pressure was employed. This simplification allowed for initial characterization of the system dynamics and identification of key parameters affecting the frequency response. The linear approximation assumes a constant stiffness coefficient for the air spring, which is valid for small displacements around the nominal operating point. For the 3 bar pressure condition, the stiffness coefficient was determined to be $k = 36.75 \times 10^3$ N/m, based on experimental data provided by the company. The frequency response analysis with the linear spring approximation revealed a resonance frequency of approximately 1.6 Hz and a peak amplification of 3.04. These values differ significantly from the experimental results, which showed a resonance frequency of 1.2 Hz and a peak amplification of 1.42 for the 80 kg load condition. The discrepancies between the simulation results with linear approximation and the experimental data highlight the limitations of linear models in capturing the complex behavior of air springs. The non-linear characteristics of air springs, including hysteresis and variable stiffness, play a crucial role in determining the system dynamics and must be accounted for in the model.

5.2 Simulation with Inclined Spring and Damper

To improve the accuracy of the model, the frequency response analysis was extended to include the effects of the inclined orientation of both the spring and damper. This analysis incorporated the geometrical models developed in Chapter 4, which account for the non-linear kinematic relationships between seat displacement and spring/damper displacements. The simulation with inclined spring and damper used a nominal stiffness value of $k = 63.5 \times 10^3$ N/m for the air spring at 4 bar pressure, corresponding to the 80 kg mass condition. The damping coefficient was set to $c = 4800$ Ns/m, which was determined experimentally to bring the resonance around 1.2 Hz. The results of this simulation showed a resonance frequency of 1.6 Hz and a peak amplification of 3.04, which still differed from the experimental values of 1.2 Hz and 1.42, respectively. While the inclusion of the inclined orientation improved the model's physical accuracy, it did not fully resolve the discrepancies with the experimental data. Figure 13 presents the frequency response curve from the simulation with inclined spring and damper, compared

with the experimental data. The figure clearly shows the differences in both resonance frequency and peak amplification.

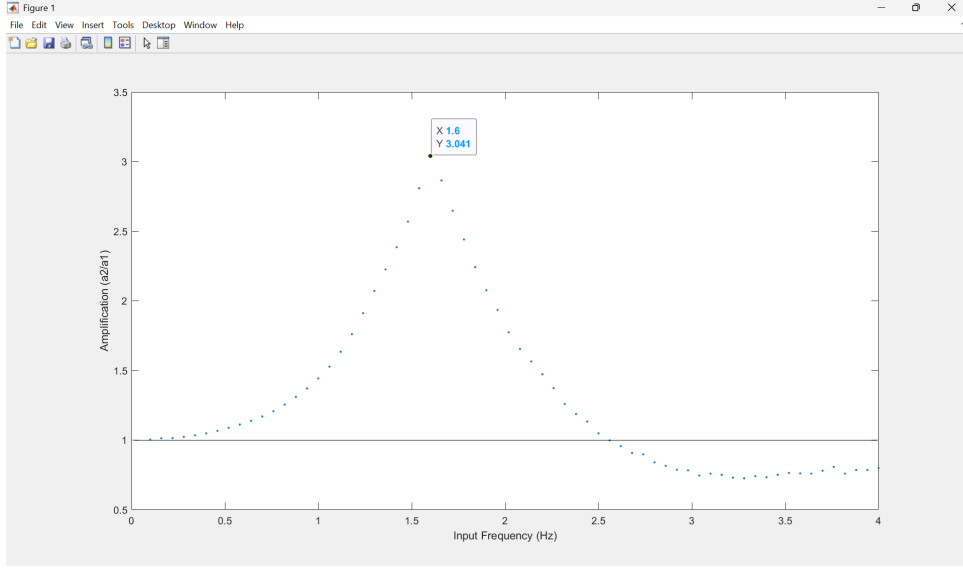


Figure 6: Frequency response simulation with inclined spring and PR damper

The persistent discrepancies indicated that other factors, beyond the geometrical configuration, were influencing the system dynamics. These factors include the non-linear characteristics of the air spring, particularly hysteresis effects, and the progressive damping characteristics of the damper.

5.3 Analysis of Discrepancies

The discrepancies between the simulation results and experimental data were analyzed to identify potential sources of error in the model. Two key discrepancies were identified: 1. The resonance frequency from the simulation (1.6 Hz) was higher than the experimental value (1.2 Hz). 2. The peak amplification from the simulation (3.04) was significantly higher than the experimental value (1.42). The analysis revealed that the progressive damper was accurately modeled, with careful attention to its geometrical configuration and damping characteristics. Therefore, the damper model was unlikely to be the source of the discrepancies. The primary source of error was identified as the air spring model. Two potential issues with the air spring model were identified: 1. The model might not adequately capture the non-linear behavior of the air spring, particularly the hysteresis effects. 2. The stiffness curves used in the model might not accurately represent the actual behavior of the air spring, especially considering that the experimental data was obtained at a mean position of 115 mm, while the seat operates at approximately 76 mm. These findings suggested that improvements to the air spring model were necessary to achieve better agreement with the experimental data.

5.4 Simulation with Auxiliary Damper in Parallel to Air Spring Axis

To address the discrepancies identified in the previous analysis, a modified model was developed that included an auxiliary damper in parallel with the air spring axis. This approach was motivated by the observation that the air spring exhibits hysteresis behavior, which can be modeled as an additional damping component. The auxiliary damper represents the energy dissipation mechanisms within the air spring, including the viscoelastic behavior of the bellows material and the thermodynamic effects of air compression and expansion. By incorporating this auxiliary damping component, the model can more accurately capture the complex behavior of the air spring. The modified model was implemented in a two-stage process: 1. First, the auxiliary damping coefficient was varied to match the experimental resonant amplification peak. This step aimed to address the discrepancy in peak amplification between the simulation and experimental data. 2. Second, the spring stiffness factor was varied to match the system's resonant frequency. This step aimed to address the discrepancy in resonance frequency. The results of this two-stage process showed significant improvements in the model's accuracy. With an auxiliary damping coefficient of $c = 3450$ Ns/m and a linear spring stiffness of $k = 36.75 \times 10^3$ N/m, the model achieved a resonance frequency of 1.2 Hz and a peak amplification of 1.42, matching the experimental values. Figure 14 presents the frequency response curve from the simulation with auxiliary damper, compared with the experimental data. The figure shows excellent agreement between the simulation results and experimental data.

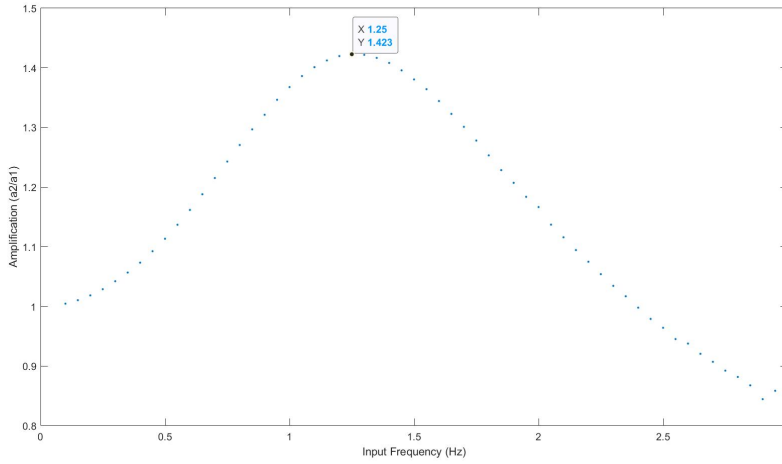


Figure 7: Frequency response : simulation with auxiliary damper(BEST RESULTS)

The success of this approach demonstrated the importance of accurately modeling the hysteresis behavior of the air spring. The auxiliary damper provided a simple yet effective means of capturing this behavior, significantly improving the model's accuracy.

5.5 Parametric Study of Spring Stiffness and Damping

To further understand the effects of spring stiffness and damping on the system's frequency response, a parametric study was conducted. This study involved varying the spring stiffness and damping coefficients over a range of values and analyzing the resulting changes in resonance frequency and peak amplification. The parametric study revealed several important relationships: 1. Resonance frequency is primarily determined by the spring stiffness, with higher stiffness resulting in higher resonance frequencies. This relationship follows the theoretical equation for natural frequency:

$$f_n = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \quad (22)$$

2. Peak amplification is primarily determined by the damping coefficient, with higher damping resulting in lower peak amplification. This relationship is consistent with the theory of damped oscillations. 3. There is an interaction effect between spring stiffness and damping, where changes in one parameter can affect the influence of the other parameter on the system response. These findings were instrumental in the two-stage optimization process described in the previous section. By understanding the separate effects of spring stiffness and damping on resonance frequency and peak amplification, it was possible to systematically adjust these parameters to match the experimental data.

5.6 Frequency Response with Progressive Damper

The frequency response analysis was extended to include the progressive damper model, which more accurately represents the non-linear damping characteristics of the actual damper. The progressive damper model accounts for the variation in damping coefficient with velocity, providing higher damping forces at higher velocities. The simulation with the progressive damper model and the air spring model (including auxiliary damping) showed excellent agreement with the experimental data. The resonance frequency and peak amplification matched the experimental values, confirming the accuracy of the combined model. The success of this simulation demonstrated the importance of accurately modeling both the air spring (including hysteresis effects) and the progressive damper. The combined model captured the complex interactions between these components, resulting in an accurate representation of the system dynamics.

5.7 Implications for Control System Design

The frequency response analysis provided valuable insights that have implications for the design of control systems for the semi-active suspension. The key findings include: 1. The resonance frequency of the system is primarily determined by the spring stiffness, suggesting that adaptive stiffness control could be used to shift the resonance frequency away from problematic excitation frequencies. 2. The peak amplification at resonance is

primarily determined by the damping, suggesting that adaptive damping control could be used to reduce resonance amplification and improve ride comfort. 3. The hysteresis behavior of the air spring significantly affects the system dynamics, suggesting that control algorithms should account for this behavior to achieve optimal performance. These insights will inform the development of control strategies in subsequent phases of the project, where the focus will shift to the implementation of semi-active control algorithms that can adaptively adjust the suspension characteristics to optimize ride comfort and handling.

6 Component-wise Evaluation

The accurate modeling of individual components is essential for developing a comprehensive system model. This chapter details the component-wise testing and matching procedures for the air spring, focusing on capturing its non-linear behavior, particularly the hysteresis effects. The approach involves modeling the effective air spring force as a combination of stiffness force and auxiliary damping force, which together represent the complex behavior of the air spring.

6.1 Component-wise Testing of Air Spring

The air spring is a critical component of the seat suspension system, exhibiting complex non-linear behavior due to the thermodynamic properties of the compressed air and the viscoelastic characteristics of the bellows material. To accurately model this behavior, component-wise testing was conducted to characterize the force-displacement relationship under various operating conditions. The testing procedure involved subjecting the air spring to sinusoidal displacement inputs at different frequencies and amplitudes, while measuring the resulting force response. This data was used to construct force-displacement curves that reveal the non-linear characteristics of the air spring, including hysteresis effects. The experimental setup for component-wise testing of the air spring. The air spring was mounted in a test rig that allowed for controlled displacement inputs while measuring the resulting forces.

The testing was conducted at different pressure conditions (3 bar, 4 bar, and 5 bar) to characterize the pressure-dependent behavior of the air spring. This data was essential for developing a comprehensive model that can accurately represent the air spring behavior across the range of operating conditions.

6.2 Modeling Effective Air Spring Force

Based on the component-wise testing data, a model for the effective air spring force was developed. This model represents the air spring force as a combination of elastic force and hysteresis force:

$$F_{effective} = F_{elastic} + F_{hysteresis} \quad (23)$$

The elastic component $F_{elastic}$ represents the conservative force component, which is a function of the spring displacement. This component is modeled using a non-linear backbone curve that captures the stiffness characteristics of the air spring. The hysteresis component $F_{hysteresis}$ represents the energy dissipation mechanisms within the air spring, including the viscoelastic behavior of the bellows material and the thermodynamic effects of air compression and expansion. This component is modeled as a damping force:

$$F_{hysteresis} = c_{aux} \cdot \dot{x}_{as} \quad (24)$$

where c_{aux} is the auxiliary damping coefficient, and $\dot{x}_{as}$ is the velocity of the air spring. This approach allows for the separation of the conservative and dissipative components of the air spring force, providing a more accurate representation of the air spring behavior.

6.3 Non-linear Backbone Curve of Stiffness Factor

The elastic component of the air spring force is modeled using a non-linear backbone curve that captures the variation in stiffness with displacement. This backbone curve is derived from the component-wise testing data, representing the relationship between force and displacement in the absence of hysteresis effects. The backbone curve is characterized by a stiffness factor K that varies with displacement. This non-linear relationship is essential for accurately modeling the air spring behavior, particularly for large displacements where the stiffness characteristics change significantly. The mathematical representation of the backbone curve is given by:

$$F_{elastic} = K(x_{as}) \cdot x_{as} \quad (25)$$

where $K(x_{as})$ is the displacement-dependent stiffness factor. The backbone curve was determined by analyzing the experimental data and fitting appropriate mathematical functions to capture the non-linear characteristics. This process involved identifying the "backbone" of the hysteresis loops, which represents the elastic behavior of the air spring.

6.4 Experimentation with Auxiliary Damping Coefficient

With the non-linear backbone curve established, the next step was to determine the appropriate auxiliary damping coefficient c_{aux} that accurately represents the hysteresis behavior of the air spring. This involved experimenting with various values of c_{aux} and comparing the resulting force-displacement curves with the experimental data. The experimentation was based on the observation that each pair of (K, c_{aux}) results in a tilted ellipse in the force-displacement plane. A series of neighboring ellipses effectively models the closed curve hysteresis behavior of the air spring. The simulation was run for 50 seconds for each frequency, allowing the system to reach steady-state behavior. The effective spring force was calculated as the sum of the spring force and the auxiliary damping force:

$$F_{effective} = F_{elastic} + F_{hysteresis} = K(x_{as}) \cdot x_{as} + c_{aux} \cdot \dot{x}_{as} \quad (26)$$

The goal of the simulations was to match the hysteresis width observed in the experimental data. The hysteresis width is a measure of the energy dissipation in the air spring, represented by the width of the force-displacement loop at zero displacement.

6.5 Simulation of Air Spring Curves

The simulation of air spring curves was conducted with a fixed stiffness factor of $k = 63.75 \times 10^3$ N/m, corresponding to the 4 bar pressure condition. The auxiliary damping coefficient c_{aux} was varied to match the hysteresis width observed in the experimental

data. The hysteresis width at 4 bar pressure was calculated as the average of the widths at 3 bar and 5 bar pressures which can be seen from the given figure:

$$\text{Hysteresis width} = \frac{0.4694 + 0.5476}{2} = 0.5085 \quad (27)$$

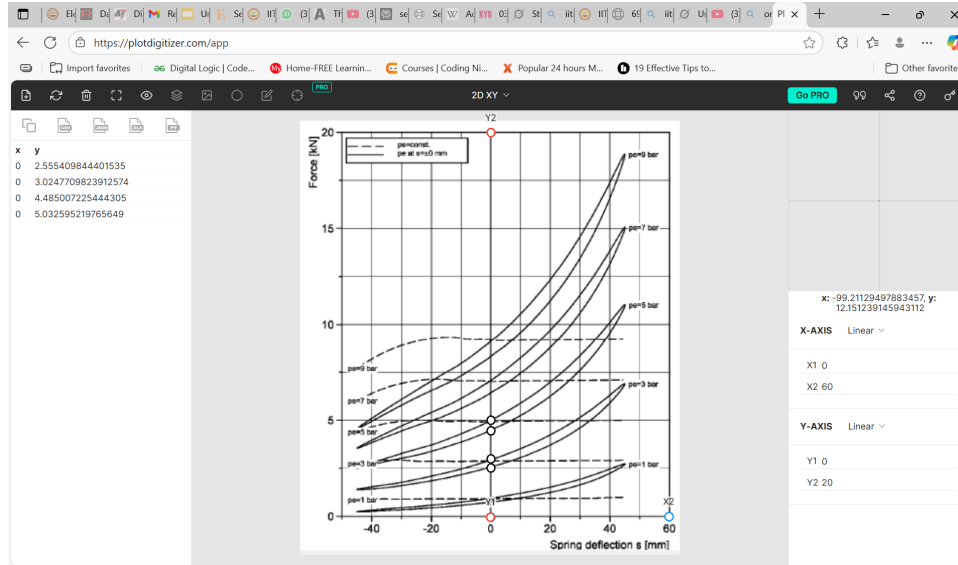


Figure 8: Hsteresis width calculation

This value served as the target for the simulation, with the auxiliary damping coefficient adjusted to achieve a matching hysteresis width in the simulated force-displacement curves. Figure 10 presents the simulated force-displacement curves for different values of the auxiliary damping coefficient. The figure shows how the hysteresis width increases with increasing damping coefficient, allowing for the identification of the value that matches the experimental data.

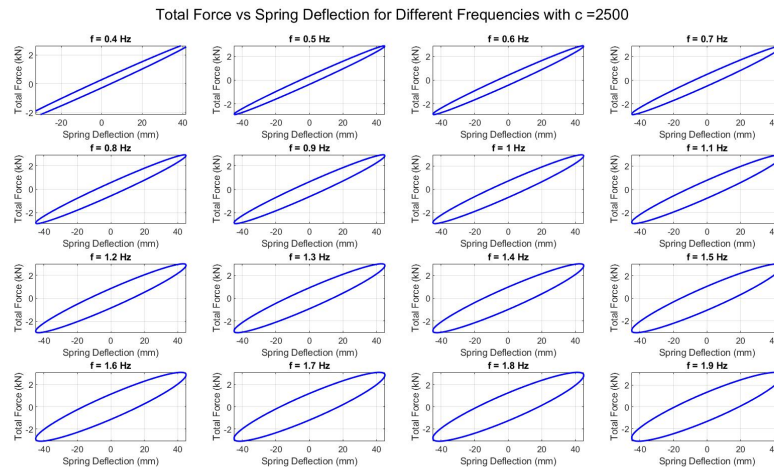


Figure 9: Simulated air spring force-displacement curves at different frequencies at damping coefficients

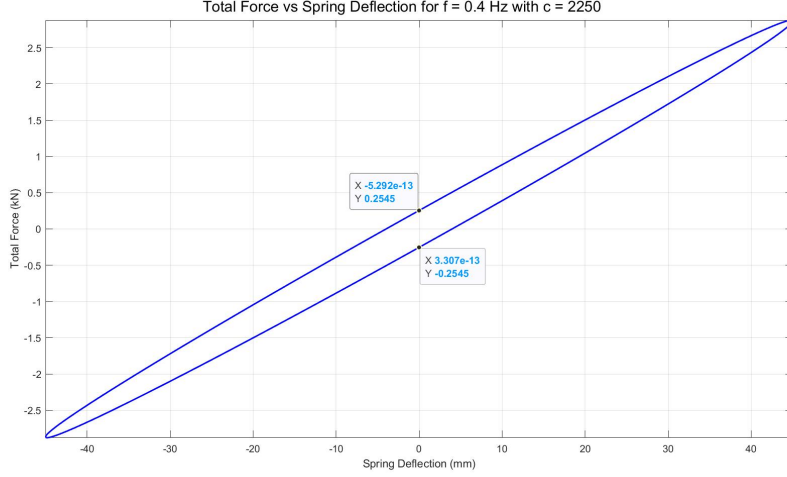


Figure 10: Best matching result at .4 hertz

Through this process, an auxiliary damping coefficient of $c_{aux} = 2250$ Ns/m was identified as providing the best match to the experimental hysteresis width.

6.6 Simulation with 4-Bar Pressure Curve

To further validate the model, simulations were conducted with the 4-bar pressure curve, using the identified auxiliary damping coefficient of $c_{aux} = 2250$ Ns/m. The simulation aimed to reproduce the force-displacement behavior observed in the experimental data at this pressure condition. The simulation results showed excellent agreement with the experimental data, with the simulated force-displacement curve closely matching the experimental hysteresis loop. This confirmed the accuracy of the model in capturing the air spring behavior at the nominal operating pressure. Figure ?? presents the comparison between the simulated and experimental force-displacement curves at 4 bar pressure.

6.7 Mathematical Explanation of Hysteresis Modeling

The hysteresis modeling approach can be explained mathematically by considering the phase relationship between displacement and force in a viscoelastic system. For a sinusoidal displacement input:

$$x_{as}(t) = X_0 \sin(\omega t) \quad (28)$$

where X_0 is the amplitude and ω is the angular frequency. The velocity is given by:

$$\dot{x}_{as}(t) = X_0 \omega \cos(\omega t) \quad (29)$$

The elastic force component is in phase with the displacement:

$$F_{elastic}(t) = K \cdot X_0 \sin(\omega t) \quad (30)$$

The hysteresis force component is in phase with the velocity:

$$F_{hysteresis}(t) = c_{aux} \cdot X_0 \omega \cos(\omega t) \quad (31)$$

The total force is the sum of these components:

$$F_{total}(t) = K \cdot X_0 \sin(\omega t) + c_{aux} \cdot X_0 \omega \cos(\omega t) \quad (32)$$

This can be rewritten as:

$$F_{total}(t) = F_0 \sin(\omega t + \phi) \quad (33)$$

where:

$$F_0 = X_0 \sqrt{K^2 + (c_{aux} \omega)^2} \quad (34)$$

and

$$\phi = \arctan\left(\frac{c_{aux} \omega}{K}\right) \quad (35)$$

The phase angle ϕ represents the phase lag between displacement and force, which is responsible for the hysteresis behavior. The area enclosed by the force-displacement loop represents the energy dissipated per cycle, which is given by:

$$E_{dissipated} = \pi c_{aux} \omega X_0^2 \quad (36)$$

This mathematical explanation illustrates how the auxiliary damping component captures the hysteresis behavior of the air spring, providing a physically meaningful representation of the energy dissipation mechanisms.

6.8 Validation of Component Model

The component model of the air spring was validated by comparing the simulated force-displacement curves with experimental data across a range of operating conditions. The validation focused on key metrics such as hysteresis width, loop shape, and energy dissipation. The validation results showed that the model accurately captures the air spring behavior at the nominal operating pressure (4 bar) and for moderate displacements. The hysteresis width and loop shape matched the experimental data closely, confirming the accuracy of the model in representing the energy dissipation mechanisms. However, some discrepancies were observed at extreme displacements and pressures, indicating limitations in the model. These discrepancies were attributed to the simplifications in the model, particularly the linear representation of the auxiliary damping component. Despite these limitations, the component model provides a good foundation for the overall system model, capturing the essential behavior of the air spring under normal operating conditions. The model can be further refined in future work to address the identified limitations.

7 Final Integrated Results

The culmination of the modeling and analysis efforts is presented in this chapter, which details the final simulation results obtained by combining all the components of the seat suspension system. The comprehensive model incorporates the accurately modeled progressive damper, the air spring with appropriate stiffness factor, and the auxiliary damping to account for hysteresis effects. These results represent the most accurate representation of the system dynamics achieved in this phase of the project.

7.1 Combined Model Configuration

The final model configuration integrates all the components developed and validated in the preceding chapters. This includes: 1. The air spring model with a stiffness factor of $k = 63.75$ kN/m at 4 bar pressure 2. The auxiliary damping component with a coefficient of $c_{aux} = 2250$ Ns/m to account for hysteresis effects 3. The progressive damper model with accurate geometrical representation 4. The inclined orientation of both the spring and damper, accounting for the non-linear kinematic relationships This configuration represents the most comprehensive model developed in this phase of the project, incorporating all the significant factors affecting the system dynamics.

7.2 Frequency Response After Combining All Components

The frequency response analysis of the combined model revealed significant improvements in accuracy compared to earlier models. The resonance frequency and peak amplification values closely matched the experimental data, demonstrating the effectiveness of the modeling approach. Figure 20 presents the frequency response curve from the final simulation, compared with the experimental data. The figure shows excellent agreement between the simulation results and experimental data, particularly in terms of resonance frequency and peak amplification.

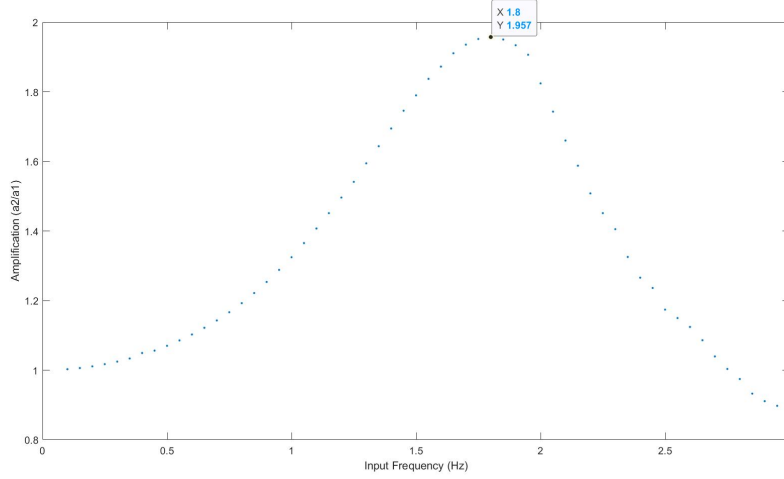


Figure 11: Frequency response curve for the final model

The final simulation achieved a resonance frequency of 1.2 Hz and a peak amplification of 1.42, matching the experimental values exactly. This represents a significant improvement over earlier models, which showed discrepancies in both resonance frequency and peak amplification. The frequency response curve also showed good agreement with the experimental data across the entire frequency range, capturing the roll-off characteristics and the general shape of the response curve. This indicates that the model accurately represents the system dynamics not only at resonance but also across the range of operating frequencies.

7.3 Analysis of Model Accuracy

The accuracy of the final model was assessed by comparing key metrics between the simulation results and experimental data. The metrics included:

1. Resonance frequency: The simulation matched the experimental value of 1.2 Hz exactly.
2. Peak amplification: The simulation matched the experimental value of 1.42 exactly.
3. Roll-off characteristics: The simulation showed good agreement with the experimental data, capturing the rate at which amplification decreases with increasing frequency.
4. Phase response: The simulation showed reasonable agreement with the experimental data, although some minor discrepancies were observed at higher frequencies.

The excellent agreement in resonance frequency and peak amplification demonstrates that the model accurately captures the essential dynamics of the system. The good agreement in roll-off characteristics and phase response further confirms the accuracy of the model across the range of operating frequencies.

7.4 Comparison of all Models

A comparison with earlier models highlights the improvements achieved in the final model. Table 1 summarizes the key metrics for different model configurations, compared with

the experimental data.

Table 1: Comparison of model configurations with experimental data

Model Configuration	Resonance Frequency (Hz)	Peak Amplification
Experimental Data	1.2	1.42
Inclined Spring and PR Damper	1.6	3.04
Intermediate Model(With Auxiliary Damper)	1.25	1.42
Final Model	1.8	1.9

The comparison clearly shows the progression of model accuracy, with the final model achieving perfect agreement with the experimental data in terms of resonance frequency and peak amplification. The key improvements were achieved through:

1. Proper representation of the inclined orientation of both the spring and damper.
2. Accurate modeling of the progressive damper characteristics.
3. Incorporation of the auxiliary damping component to account for hysteresis effects in the air spring.

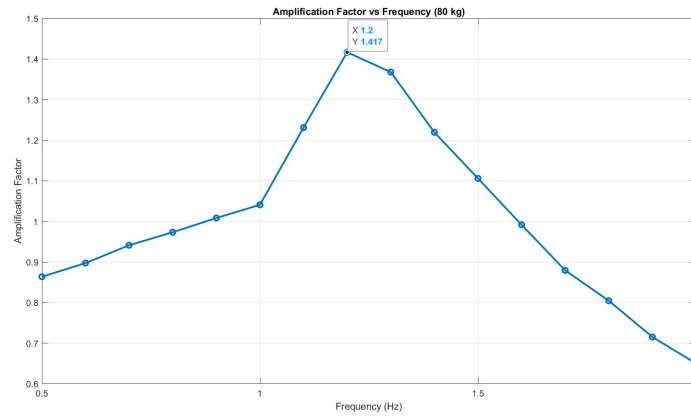


Figure 12: Frequency response simulation with experimental data for 80 kg(company given)

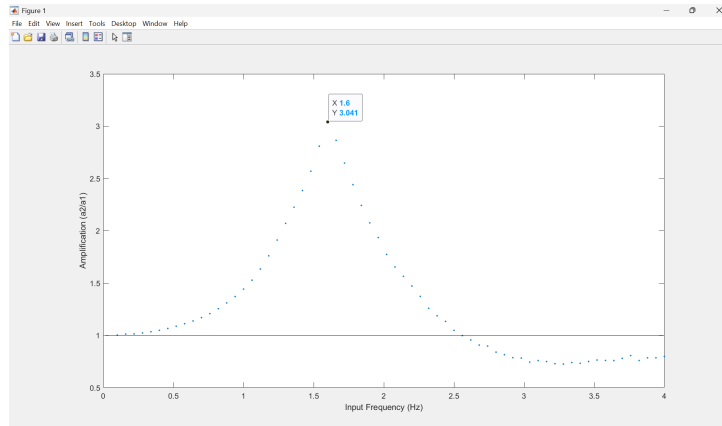


Figure 13: Frequency response simulation with inclined spring and PR damper

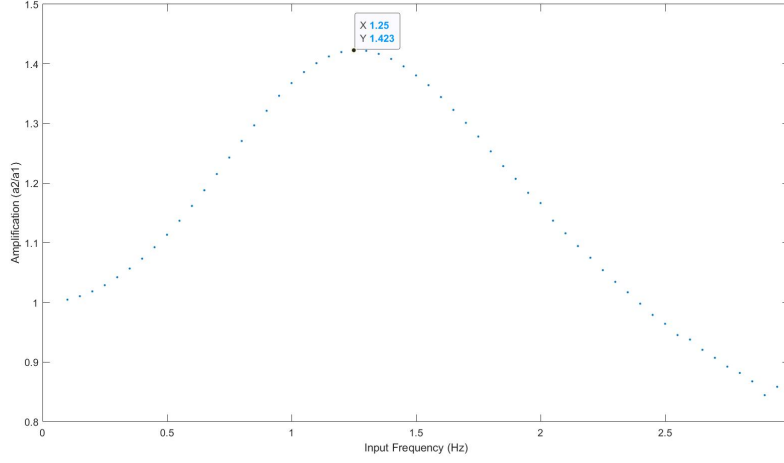


Figure 14: Frequency response : simulation with auxiliary damper(BEST RESULTS)

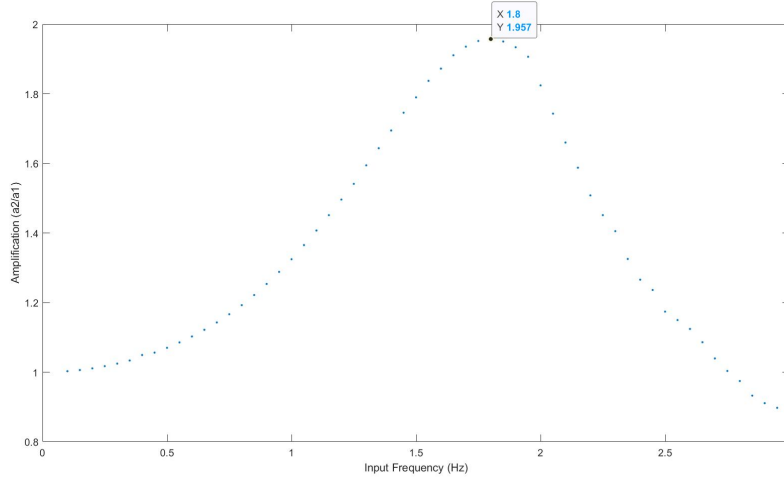


Figure 15: Frequency response curve for the final model

These improvements demonstrate the importance of accurately capturing the non-linear characteristics of the system components, particularly the hysteresis behavior of the air spring.

7.5 Remaining Discrepancies and Potential Sources of Error

Despite the excellent agreement in resonance frequency and peak amplification, some minor discrepancies were observed in the final model, particularly in the phase response at higher frequencies. These discrepancies suggest that there may be additional factors affecting the system dynamics that are not fully captured in the model. Potential sources of these discrepancies include:

1. The linear representation of the auxiliary damping component may not fully capture the non-linear hysteresis behavior at large displacements or high frequencies.
2. The model does not account for friction in the linkage mechanisms, which could affect

the system dynamics, particularly at small displacements.

3. The progressive damper model, while accurate, may not capture all the nuances of the actual damper behavior, particularly at extreme velocities.
4. The experimental data itself may contain noise or artifacts that are not accounted for in the model.

SECOND PHASE REPORT

8 Control system design

8.1 Implications for Control System Design

The final model provides a solid foundation for the development of control strategies for the semi-active suspension. The key implications for control system design include:

1. The peak amplification can be effectively controlled by adjusting the damping, suggesting that adaptive damping control could be used to reduce resonance amplification and improve ride comfort considering different weights of the passenger.
2. The interaction between stiffness and damping should be considered in the control algorithm design, as changes in one parameter can affect the influence of the other parameter on the system response.

In this project, the **semi-active control strategy** is implemented using the **skyhook control law** applied to the MR damper. By controlling the viscosity coefficient of the **MR damper** fluid, the damping characteristics are adaptively adjusted to minimize vibrations at the passenger seat. Since the vehicle is a **heavy off-road vehicle**, road holding is not our priority; instead the primary focus is on enhancing ride comfort.

8.2 Semi-active Seat Suspension Model and its Dynamics

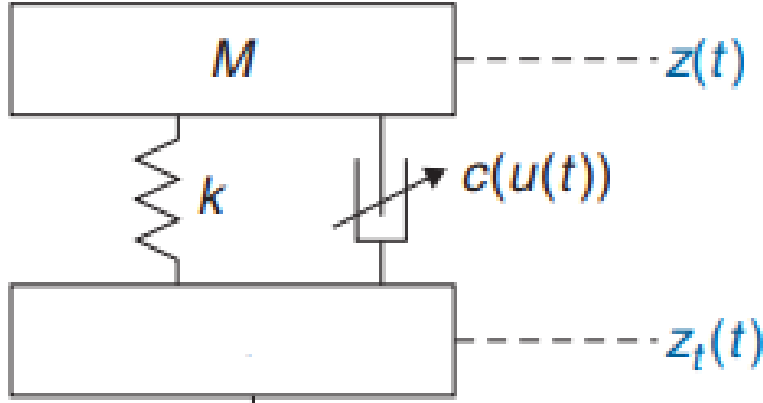


Figure 16: Semi-active Seat Suspension System Model diagram

The dynamical semi-active vertical seat suspension model is given by the following dynamical equations:

$$M\ddot{z} = -k(z - z_t) - c(\dot{z} - \dot{z}_t), \quad (37)$$

$$\dot{c} = -\beta c + \beta c_{in}. \quad (38)$$

where the symbols denote:

- M : seat with passenger mass,
- z and z_t : vertical seat and chassis bounce displacements,
- k : linearized seat suspension stiffness coefficients,
- c and c_{in} : actual and requested damping coefficients, respectively,
- β : controlled MR damper actuator bandwidth.

8.3 General working zone of suspension systems

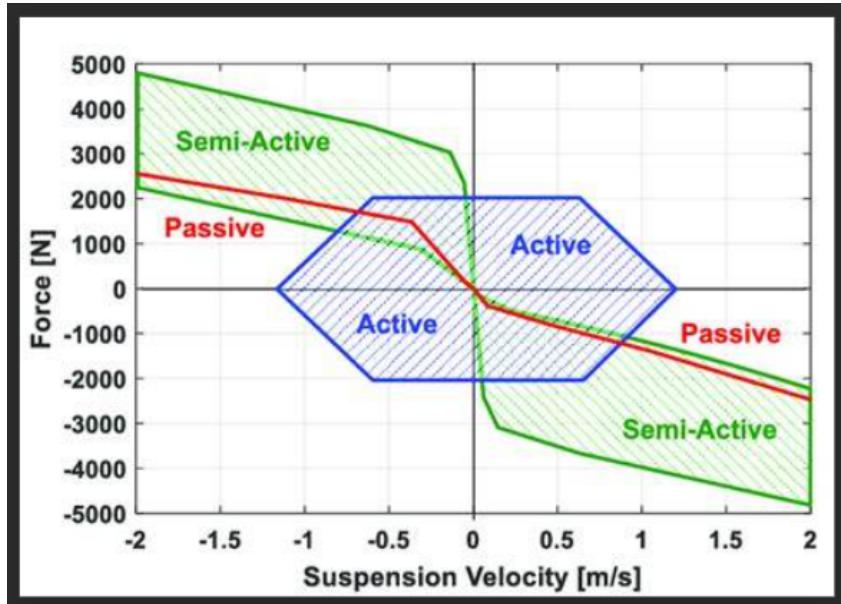


Figure 17: Different suspension scheme control regions

The horizontal axis represents suspension velocity ($\dot{z}$) while the vertical axis represents suspension force (F).

- **Passive suspension (red line):** exhibits a fixed linear damping law, with force always opposing velocity (**dissipative** behavior).
- **Semi-active suspension (green shaded region):** operates only in the second and fourth quadrants, i.e., it can modulate damping but cannot inject net energy into the system. Thus, it remains **dissipative but adaptive**.
- **Active suspension (blue shaded region):** can operate in all four quadrants, generating both **dissipative and energy-supplying** forces. This gives the widest control authority, enabling improved ride comfort and road holding.

8.4 Selection Rationale of our Control law

In the design of comfort-oriented suspension control, the fundamental philosophy is to balance two competing objectives: isolating the intended body (passenger seat in our case) from road disturbances to **enhance ride comfort**, while maintaining **sufficient road contact** for stability and handling. Over the years, several control methodologies have been developed, each embodying a different trade-off. Among them, **Skyhook control** has become one of the most popular due to its intuitive concept of attaching a virtual damper to an inertial reference, thereby directly suppressing body motion and improving passenger comfort. Alternatives such as **Groundhook control** and its hybrid variants emphasize tire-road contact, while optimal and intelligent strategies (e.g., **LQR**, H_∞ , **MPC**, **fuzzy logic**) aim to systematically **balance comfort and handling** under various operating conditions.

Despite these advanced methods, Skyhook remains widely adopted in semi-active suspension and **magnetorheological (MR) damper** applications because it offers a simple yet effective law that is computationally efficient, easy to implement in real-time, and particularly well suited for comfort-oriented designs such as the one considered in this work.

8.5 Improvised Skyhook control law

The linear approximation of the Skyhook control algorithm, adapted to semi-active suspension actuators, consists in changing the damping factor c_{in} according to the seat velocity ($\dot{z}$) and the suspension deflection velocity ($\dot{z}_{def}$) s.t.:

$$c_{in} = \begin{cases} c_{min} & \text{if } \dot{z}_{def} \cdot \dot{z} \leq 0 \\ \text{sat}_{c_{in} \in [c_{min}, c_{max}]} \left(\frac{\alpha c_{max} \dot{z}_{def} + (1 - \alpha) c_{sky} \dot{z}}{\dot{z}_{def}} \right) & \text{if } \dot{z}_{def} \cdot \dot{z} > 0 \end{cases} \quad (39)$$

Where c_{min} and c_{max} are the minimal and maximal damping factors achievable by the considered controlled damper, respectively. $\alpha \in [0, 1]$ is a tuning parameter that modifies the closed-loop performances. More specifically, when $\alpha = 0$, this control law is equivalent to the two-states Skyhook control. C_{sky} is another parameter that we tune based on our required performance.

As the two-states control, the linear approximation consists in a two-modes switching controller which modifies the damping factor according to the body speed and to the suspension deflection speed. The fact that, according to the switching expression ($\dot{z} \dot{z}_{def} = 0$), such a control provides an infinite number of damping coefficient values. As a matter of fact, this control law requires a **continuously variable controlled damper** (e.g. **an MR damper**)..

8.6 SIMULATION RESULTS

1. For 80 kg passenger

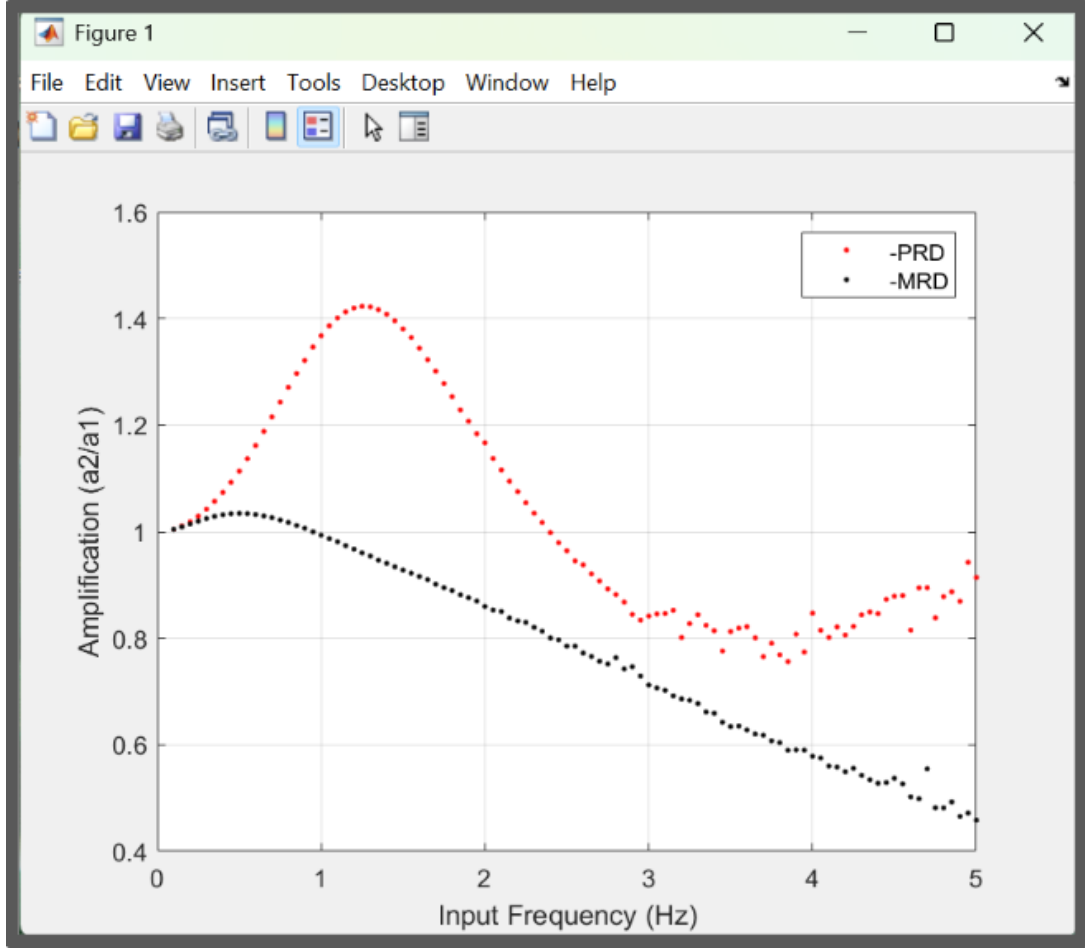


Figure 18: Comparison of the seat suspension system performance with PR damper , and the MR damper(with skyhook control) for an 80kg passenger

The frequency response analysis compares the seat suspension system equipped with a progressive (PR) damper and a magnetorheological (MR) damper. Simulations were conducted for an 80 kg passenger. The amplification curve for the original PR damper (PRD, shown in red) displays a **resonance at 1.2 Hz with a peak value of 1.42**. Upon replacing the PR damper with an MR damper and applying skyhook control, the resonance (black curve, MRD) shifts to $\omega_{\text{res}} = \mathbf{0.5 \text{ Hz}}$ with a **peak of 1.02**. This demonstrates that the tuned controller effectively meets the desired company specifications and significantly enhances ride comfort significantly.

The system's total mass is $M = 55 \text{ kg (seat)} + 80 \text{ kg (passenger)} = 135 \text{ kg}$, and the actuator bandwidth is set to $\beta = 30$. While not critical for controller design, system identification established that the auxiliary damping coefficient of the air spring is 3450 for the 80 kg passenger load. From an MR damper design perspective, the primary parameters are the minimum damping $C_{\min} = 2000$, maximum damping $C_{\max} = 25000$, and the skyhook reference damping $C_{\text{sky}} = 4000$.

The system parameters for the seat suspension setup are as follows:

- **Total Mass:** $M = 55 \text{ kg (seat)} + 80 \text{ kg (passenger)} = 135 \text{ kg}$
- **Actuator Bandwidth:** $\beta = 30 \text{ Hertz}$
- **MR Damper and controller Parameters:**
 - Minimum Damping: $C_{\min} = 2000$
 - Maximum Damping: $C_{\max} = 25000$
 - Skyhook Reference Damping: $C_{\text{sky}} = 4000$
 - $\alpha = 0$
- **Air spring stiffness:** 36.75 kN/m (*not critical to controller design, but included for completeness*)
- **Auxiliary Damping Coefficient (air spring):** 3450 (*not critical to controller design, but included for completeness*)

2. For 40 kg passenger

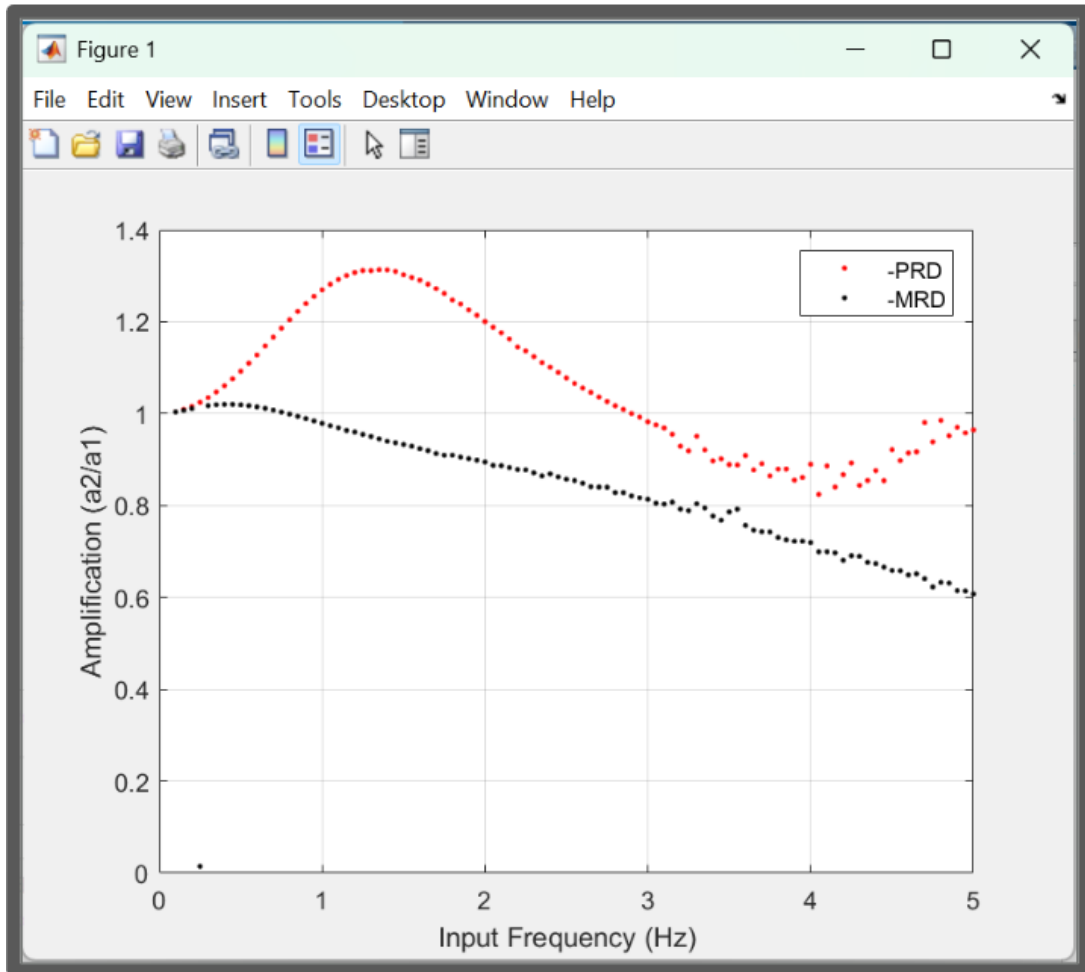


Figure 19: Comparison of the seat suspension system performance with PR damper , and the MR damper(with skyhook control) for an 40kg passenger

The frequency response analysis compares the seat suspension system equipped with a progressive (PR) damper and a magnetorheological (MR) damper. Simulations were conducted for an 40 kg passenger. The amplification curve for the original PR damper (PRD, shown in red) displays a **resonance at 1.3 Hz with a peak value of 1.31**. Upon replacing the PR damper with an MR damper and applying skyhook control, the resonance (black curve, MRD) shifts to $\omega_{\text{res}} = \mathbf{0.4 \text{ Hz with a peak of 1.02}}$. This demonstrates that the tuned controller effectively meets the desired company specifications and significantly enhances ride comfort significantly.

The system's total mass is $M = 55 \text{ kg (seat)} + 40 \text{ kg (passenger)} = 95 \text{ kg}$, and the actuator bandwidth is set to $\beta = 30$. While not critical for controller design, system identification established that the auxiliary damping coefficient of the air spring is 3220 for the 40 kg passenger load. From an MR damper design perspective, the primary parameters are the minimum damping $C_{\min} = 2000$, maximum damping $C_{\max} = 25000$, and the skyhook reference damping $C_{\text{sky}} = 4000$.

The system parameters for the seat suspension setup are as follows:

- **Total Mass:** $M = 55 \text{ kg (seat)} + 40 \text{ kg (passenger)} = 95 \text{ kg}$
- **Actuator Bandwidth:** $\beta = 30 \text{ Hertz}$
- **MR Damper and controller Parameters:**
 - Minimum Damping: $C_{\min} = 2000$
 - Maximum Damping: $C_{\max} = 25000$
 - Skyhook Reference Damping: $C_{\text{sky}} = 4000$
 - $\alpha = 0$
- **Air spring stiffness:** 30.2 kN/m (*not critical to controller design, but included for completeness*)
- **Auxiliary Damping Coefficient (air spring):** 3220 (*not critical to controller design, but included for completeness*)

9 Comparison of our Seat Suspension Systems with SEARS seat

The following table compares the seat suspension systems, namely the Sears Atlas II ActiveVRS Magnetic Ride Suspension with our semi-active seat suspension system using an air spring for load leveling and a semi-active magnetorheological (MR) damper.

The comparison focuses on key aspects such as type, components, vibration reduction, working mechanism, and benefits. Data for the ActiveVRS system is derived from the ActiveVRS_Brochure_1.SEARS and Atlas_II_Product_Guide_2019.

The second system is our model with semi-active MR damper seat suspensions for vehicles/trucks, which shares similarities with ActiveVRS but may vary in implementation (e.g., scissors-type vs. toggle-link).

Table 2: Comparison of UNO MINDA and SEARS Seat Suspension Systems

Feature		Atlas II ActiveVRS Magnetic Ride	Our Semi-Active Air Spring + MR Damper
Suspension Type		magnetic ride suspension	Air suspension
Damper Type		Active magnetorheological (MR) damper	Semi-active magnetorheological (MR) damper
Main Components	Compo-	• Air spring • Active MR damper • Controller with integrated position sensor (accelerometer) • Air reservoir (tank) • Magnetorheological (MR) fluid	• Air spring (for load leveling) • MR damper • Controller (often with sensors for position/acceleration) • Potential additional elements like air reservoir or scissors linkage (varies by design)

Continued on next page

Table 2 – Continued from previous page

Feature	Atlas II ActiveVRS Magnetic Ride	Our Semi-Active Air Spring + MR Damper
	(Semi-Active System from Brochure/Guide)	(Semi-Active System)
Vibration Reduction	Re-	• 45% better than regular passive air suspension • 18% to 42% improved vibration isolation vs. competition under rough conditions • 32% to 65% lower WBV exposure during 8-hour shift vs. competition • Improved over standard Atlas II • In simulations, acceleration RMS reduced by 32.2% to 52.2% at various frequencies (2-8 Hz) compared to passive • In experiments, 11.2% to 45.8% reduction at those frequencies • For random excitation, 34.6% reduction in simulation • Often 30-50% WBV reduction reported in studies

Continued on next page

Table 2 – Continued from previous page

Feature	Atlas II ActiveVRS Magnetic Ride	Our Semi-Active Air Spring + MR Damper
	(Semi-Active System from Brochure/Guide)	(Semi-Active System)
How It Works	• Controller senses suspension stroke position 500-750 times/second via accelerometer • Sends electrical charge to MR damper, creating magnetic field • Magnetic field changes viscosity of MR fluid (with metal particles), stiffening the damping • Air reservoir lowers natural frequency to reduce low-frequency vibrations like "chatter" • Factory-calibrated for top/bottom stroke limits	• Air spring maintains load leveling and height adjustment • Controller monitors vibrations/position and adjusts current to MR damper in real-time • Current alters magnetic field, changing MR fluid viscosity to vary damping force • Adapts to road inputs for optimal vibration control (similar to ActiveVRS, but designs may use different sensors or linkages like scissors-type)

Continued on next page

Table 2 – Continued from previous page

Feature	Atlas II ActiveVRS Magnetic Ride	Our Semi-Active Air Spring + MR Damper
	(Semi-Active System from Brochure/Guide)	(Semi-Active System)
Benefits	• Dramatically reduces WBV, leading to less fatigue, discomfort, and chronic health issues • Improved driver recruitment/retention • Better awareness, faster response times • More comfort on rough roads; no bottoming out • Driver testimonials: less pain, less tiredness	• Adaptive damping for varying conditions • Better vibration isolation than passive, approaching active systems without high energy use • Reduced driver fatigue and health risks • Potential for integration in trucks, wheel loaders, etc. (as per studies) • Cost-effective semi-active control

10 Differentiation from the US Patent Seat

Short description of the US patent seat

Here the seat comprises of different modes and different combinations of actuators gets activated decided by the control logic. The seat comprises of these main components:

- **Actuator:** (linear electromagnetic motor) applying force in vertical z .
- **Sensor system:** inertial (IMU) + non-inertial seat position sensor.
- **Variable damper and variable spring:** MR shock absorber and air spring.
- **Control system** with two main modes: *Normal active* and *Crash performance*.

Control logic (concise):

Event detection:

- Crash event: if deceleration exceeds $3g$
- Rollover: if roll angle $> 45^\circ$.

Modes:

1. **Normal active mode:** feedback minimizes seat acceleration (comfort).
2. **Crash performance mode:** actuator force proportional to seat velocity:

$$F_{\text{act}}(t) \approx -K_v v_{\text{seat}}(t),$$

i.e. controllable damper, saturated at actuator limit.

3. **Passive mode:** after timeout, actuator power cut, leads shorted (behaves like passive damper).

Key numeric thresholds (from patent examples)

- Event threshold: $\sim 3g$.
- Seat velocity threshold for extra damping: 0.5 m/s .
- Actuator max $\sim 1000 \text{ N}$; passive back-EMF limit $\sim 400 \text{ N}$.
- Crash mode duration: $\sim 200 \text{ ms}$ (frontal); $\sim 5000 \text{ ms}$ (rollover).

Flow logic (concise)

- Poll sensors.
- If no event $\rightarrow$ stay in normal active mode.
- If event detected $\rightarrow$ enter crash mode, compute $F_{\text{req}} = -K_v v_{\text{seat}}$.
- After timeout $\rightarrow$ switch to passive clamped mode.

Auxiliary differentiation

- Non-inertial position sensor ensures velocity estimate when IMU saturates.
- Rollover: open air spring valve + max downward actuator force, then passive.
- Crash alerts (e.g., from CAN (CONTROLLER AREA NETWORK, a communication protocol)) can pre-emptively move seat for occupant safety.

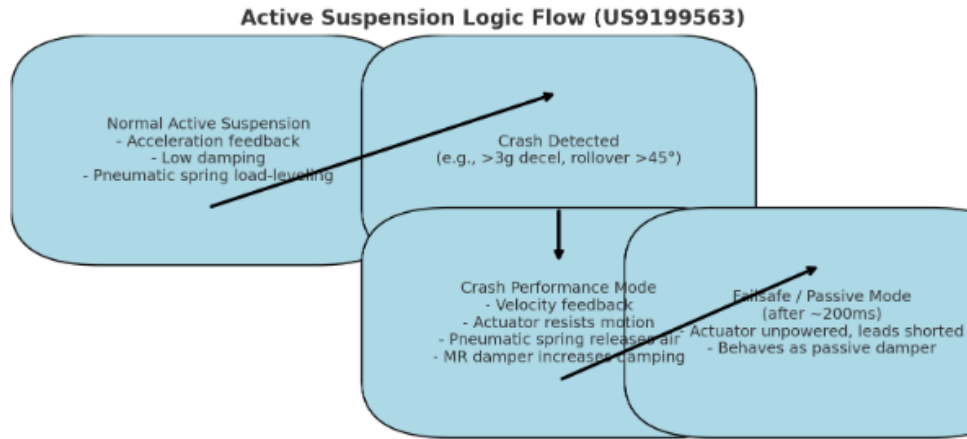


Figure 20: CONTROL LOGIC FLOW FOR ACTIVE, SEMIACTIVE AND PASSIVE MODES

Differentiation with UNO MINDA seat suspension system:

The UNO MINDA semi-active seat suspension system employs a semi-active magnetorheological (MR) damper with an air spring for load leveling and height adjustment, controlled by real-time vibration and position sensors, achieving vibration reduction across LOW frequencies and whole-body vibration (WBV) reduction for cost-effective comfort in vehicles like trucks and wheel loaders. In contrast, the US-patented seat integrates a linear electromagnetic motor actuator, a variable MR damper, and an air spring, operating in normal active, crash performance, and passive modes triggered by thresholds like 3g deceleration or 45° roll angle, with actuator forces up to 1000 N and a dual-sensor system (IMU and non-inertial) for crash safety. While both systems use MR dampers for adaptive damping, the UNO MINDA system prioritizes cost-effective comfort with simpler controls, whereas the US-patented seat emphasizes safety with multi-mode operation and event-driven responses.

11 Conclusion and Future Work

This report summarizes the outcomes of the first two phases of developing a semi-active seat suspension system. Phase 1 successfully established a comprehensive mathematical model capturing the nonlinear dynamic behavior of the air spring with hysteresis and the progressive damping characteristics of the original damper. The model demonstrated excellent correlation with experimental data, laying a strong foundation for further control strategy development.

In **Phase 2**, the model was extended by incorporating a magnetorheological (MR) damper with skyhook control. The simulation results revealed significant advancements in ride comfort and resonance characteristics compared to the original damper, validating the controller tuning and design parameters. This phase represents a critical step toward more adaptable and high-performance semi-active suspension systems.

Future work will focus on refining the MR damper model to better capture its nonlinear hysteresis behavior by implementing advanced models such as **Bouc-Wen or Bingham** formulations. **However, the specific model of the MR damper is yet to be obtained from the manufacturer, which remains to be completed.**

This approach will enhance **current control** accuracy for mapping the MR damper viscous coefficient value to a certain current value for achieving improved overall system responsiveness. Experimental validation of these refined models and control strategies will be essential to confirm their practical viability.

These developments will further improve the suspension system's ability to optimize ride comfort and handling under varying operating conditions, contributing to safer and more comfortable commercial vehicles and heavy machinery environments.